

Insurance Claims Risk Analysis & Actuarial Pricing Framework

Final Project Report

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Tools & Technologies

- R Programming
 - Microsoft Excel
 - Microsoft Power BI
 - Git & GitHub
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Project Duration

Jun 2026 – July 2026

GitHub Repository

[GITHUB](#)

Project Objective

To analyze insurance claims data using actuarial and data analytics techniques to identify key risk drivers, evaluate claim frequency and severity, develop a rule-based risk segmentation framework, and propose risk-based premium recommendations supported by an interactive Power BI dashboard.

Executive Summary

Insurance companies rely on historical claims data to understand policyholder risk, estimate future losses, and determine appropriate insurance premiums. Accurate risk assessment is essential for ensuring fair pricing, effective underwriting, and long-term financial sustainability. This project applies actuarial principles and data analytics techniques to evaluate insurance claim behaviour and develop a practical risk-based pricing framework.

The analysis was conducted using a publicly available insurance claims dataset containing **1,000 policy records** and **39 variables** related to policyholder demographics, policy details, incident characteristics, fraud status, and claim payments. The project utilized **R** for data cleaning, exploratory data analysis, statistical summaries, and visualization, **Microsoft Excel** for data validation, and **Microsoft Power BI** for interactive dashboard development.

The analytical process included exploratory data analysis, claim frequency analysis, claim severity analysis, fraud assessment, and the construction of a rule-based risk segmentation framework. Policyholders were classified into five risk categories—Very Low Risk, Low Risk, Medium Risk, High Risk, and Very High Risk—based on claim-related characteristics such as incident severity, fraud status, and the number of vehicles involved.

Using the results of the risk segmentation model, a risk-based pricing framework was developed to recommend premium adjustments that better reflect historical claim experience. The project demonstrates how differentiated pricing can improve fairness by aligning premiums with the expected level of insurance risk.

An interactive Power BI dashboard was created to summarize the analytical findings through key performance indicators, claim frequency and severity visualizations, fraud analysis, risk category distribution, and premium comparisons. The dashboard provides a user-friendly interface for exploring the insurance portfolio and supports data-driven underwriting and pricing decisions.

Overall, this project illustrates how actuarial thinking, statistical analysis, and business intelligence tools can be combined to transform historical insurance data into actionable insights for insurance portfolio management and pricing.

Project Overview

1.1 Project Objectives

The specific objectives are:

- Analyze historical insurance claims data using statistical techniques.
 - Identify key drivers influencing claim frequency and claim severity.
 - Examine fraud patterns and their impact on insurance claims.
 - Develop a five-level rule-based risk segmentation framework.
 - Propose a risk-based premium recommendation model.
 - Design an interactive Power BI dashboard for portfolio monitoring and business reporting.
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1.2 Dataset Description

Attribute	Description
Source	Kaggle – Insurance Fraud Detection Dataset
Total Records	1,000
Total Variables	39
Type	Motor Insurance Claims
Format	CSV

Key Variables

- Policy Annual Premium
 - Policy Deductible
 - Total Claim Amount
 - Incident Severity
 - Incident Type
 - Number of Vehicles Involved
 - Fraud Reported
 - Vehicle Claim
 - Property Claim
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1.3 Software and Tools

The following software tools were used throughout the project:

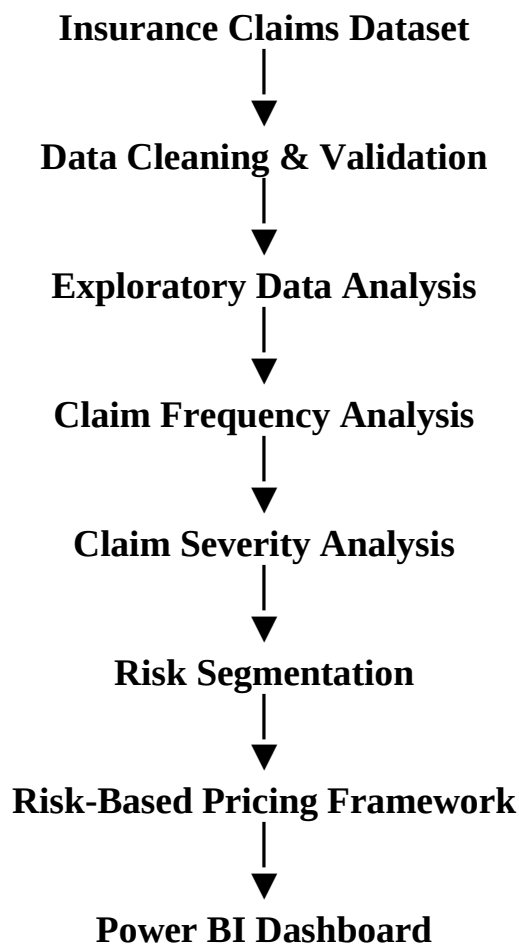
Tool	Purpose
R Programming	Data cleaning, statistical analysis, visualization
Microsoft Excel	Data validation and preprocessing
Microsoft Power BI	Dashboard development
Git & GitHub	Version control and project documentation

Methodology

2.1 Methodological Framework

This project follows a structured actuarial and data analytics workflow to transform raw insurance claims data into meaningful business insights and pricing recommendations. The methodology combines exploratory data analysis, statistical techniques, risk segmentation, and business intelligence visualization.

The overall analytical process consisted of the following stages:



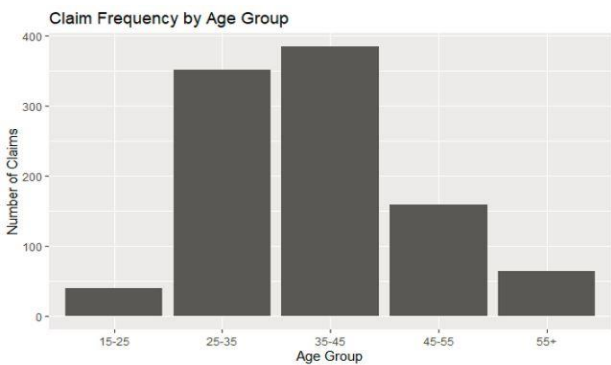
Exploratory Data Analysis

3.1 Overview

Exploratory Data Analysis (EDA) was conducted to understand the characteristics of the insurance portfolio before performing detailed actuarial analyses. The objective was to identify patterns in claim behaviour, demographic composition, and claim-related variables that could influence insurance risk.

Portfolio Summary

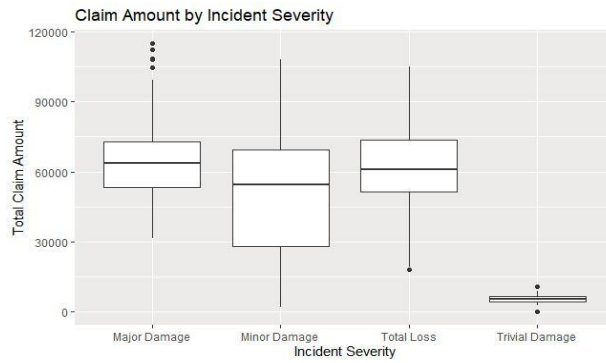
Metric	Value
Total Claims	1,000
Variables	39
Average Claim Amount	₹52,762
Minimum Claim	₹100
Maximum Claim	₹114,920



(Figure: Claim Frequency by Age Group)

Key Findings

- The **35–45 years** age group represented the largest proportion of policyholders.
- The portfolio is primarily composed of middle-aged policyholders.
- Claim amounts exhibited substantial variation, indicating differences in claim severity across the portfolio.
- Existing insurance premiums showed very little correlation with historical claim costs, suggesting opportunities for more risk-sensitive pricing.



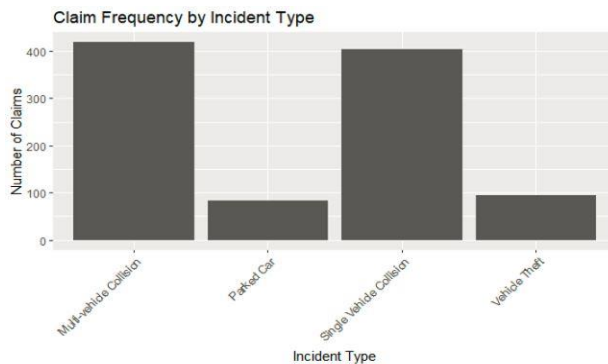
(Figure: Claim Amount by Incident Severity)

The exploratory analysis indicates that incident characteristics play a more significant role in determining claim costs than demographic variables alone.

Claim Frequency Analysis

4.1 Overview

Claim frequency analysis examines how often insurance claims occur across different policyholder and incident characteristics. Understanding claim frequency enables insurers to identify the segments contributing the highest number of claims and supports effective underwriting and portfolio management.

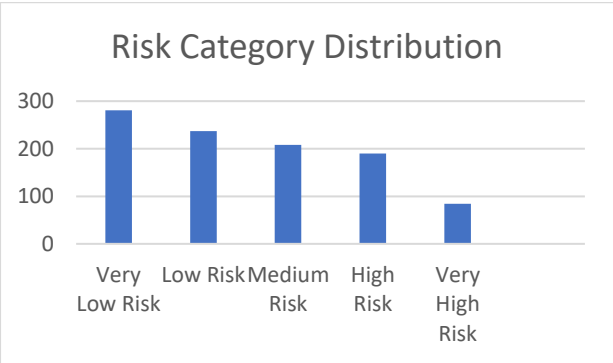


(Figure: Claim Frequency by Incident Type)

Claim Frequency by Incident Type

Incident Type	Number of Claims
Multi-vehicle Collision	419
Single Vehicle Collision	403
Vehicle Theft	94
Parked Car	84

The analysis shows that **collision-related incidents accounted for over 80% of all claims**, making them the dominant source of insurance losses within the portfolio.



(Figure: Risk Category Distribution)

Risk Category Distribution

Risk Category	Policyholders
Very Low Risk	281
Low Risk	237
Medium Risk	208
High Risk	190
Very High Risk	84

Key Findings

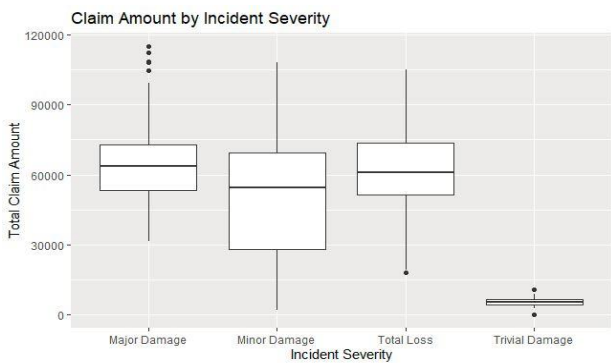
- Collision-related incidents dominate the portfolio.
- Approximately **25% of claims** were identified as fraudulent.
- Minor Damage incidents occurred most frequently.
- Most policyholders belonged to the **Very Low Risk** and **Low Risk** categories.
- High-risk policyholders represented a relatively small proportion of the portfolio but require greater underwriting attention.

These findings help insurers understand where claims occur most frequently and provide a basis for analysing their financial impact.

Claim Severity Analysis

5.1 Overview

While claim frequency analysis explains how often claims occur, claim severity analysis evaluates their financial impact. The objective is to identify policyholder and incident characteristics associated with higher insurance losses.



(Figure: Claim Amount by Incident Severity)

Average Claim Amount by Incident Severity

Incident Severity	Average Claim (₹)
Major Damage	64,067
Total Loss	62,081
Minor Damage	48,643
Trivial Damage	5,302

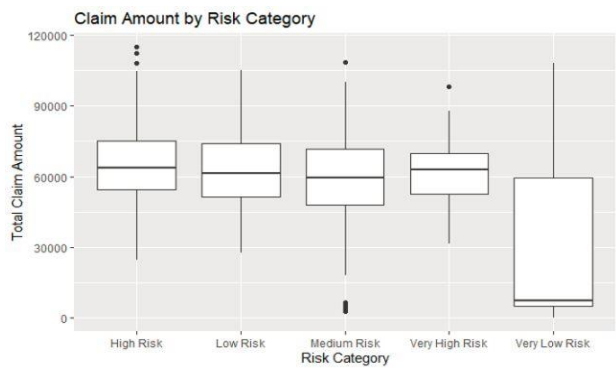
Major Damage and Total Loss incidents generated the highest average claim amounts, while Trivial Damage incidents resulted in comparatively small claims.

Fraudulent claims exhibited higher average claim amounts than genuine claims, indicating that fraud contributes significantly to insurance losses.

Key Findings

- Major Damage produced the highest claim costs.
- Fraudulent claims were more expensive than non-fraudulent claims.
- Claim severity varied substantially across incident types.
- Incident severity emerged as one of the strongest predictors of insurance losses.

Claim Severity by Risk Category

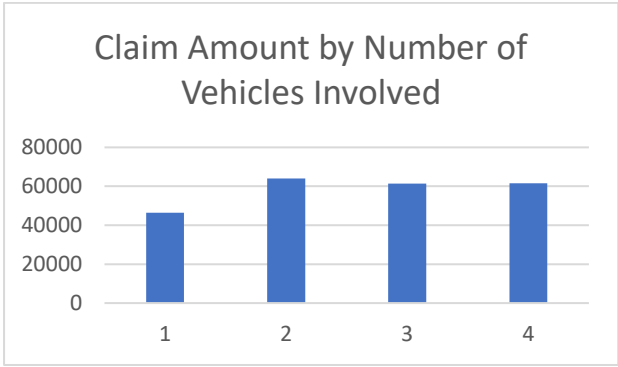


(Figure: Claim Amount by Risk Category)

Average Claim Amount by Risk Category

Risk Category	Average Claim (₹)
Very Low Risk	30,266
Low Risk	62,163
Medium Risk	57,523
High Risk	65,463
Very High Risk	60,975

The results indicate a clear distinction between the **Very Low Risk** category and the remaining policyholder groups. High Risk policyholders recorded the highest average claim amount, confirming the effectiveness of the rule-based risk segmentation framework.



(Figure: Claim Amount by Number of Vehicles Involved)

Claim Severity by Number of Vehicles Involved

Vehicles Involved	Average Claim (₹)
1	46,361
2	64,053
3	61,443
4	61,540

Incidents involving multiple vehicles generally resulted in higher claim amounts than single-vehicle accidents, reflecting the greater complexity and financial impact of such events.

Business Insights

- Higher claim severity is strongly associated with severe incidents and fraudulent claims.
- Multi-vehicle collisions contribute significantly to insurance losses.
- The five-level risk segmentation model effectively distinguishes policyholders according to expected claim costs.
- These findings provide a strong foundation for developing a differentiated pricing strategy based on historical claims experience.

Risk Segmentation & Pricing Framework

6.1 Risk Segmentation

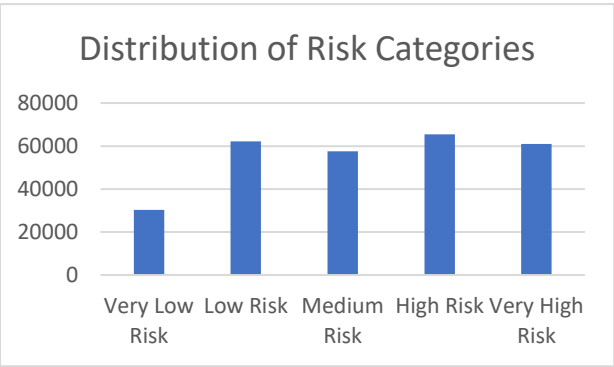
To support actuarial pricing, a rule-based risk segmentation framework was developed using three key claim characteristics:

- Incident Severity
- Fraud Status
- Number of Vehicles Involved

Each policyholder was assigned a risk score and classified into one of five categories.

Risk Categories

Risk Category	Number of Policyholders
Very Low Risk	281
Low Risk	237
Medium Risk	208
High Risk	190
Very High Risk	84



(Figure: Distribution of Risk Categories)

The majority of policyholders were classified as **Very Low Risk** and **Low Risk**, while only **8.4%** belonged to the **Very High Risk** category.

6.2 Average Claim Amount by Risk Category

Risk Category	Average Claim Amount (₹)
Very Low Risk	30,266
Low Risk	62,163
Medium Risk	57,523
High Risk	65,463
Very High Risk	60,975

The substantial variation in claim amounts across the five categories demonstrates that the segmentation model effectively differentiates policyholders according to historical claim experience.

Key Findings

- Very Low Risk policyholders generated the smallest claim costs.
- High Risk policyholders recorded the highest average claim amount.
- The framework successfully identifies higher-risk policyholders for differentiated underwriting and pricing.
- Risk segmentation provides a practical foundation for actuarial pricing decisions.

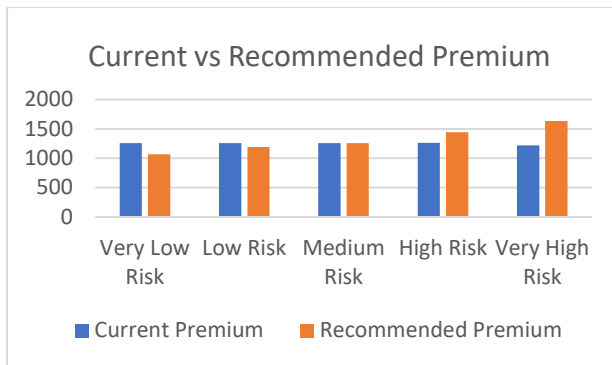
Risk-Based Pricing Framework

Using the historical claims experience of each risk category, a risk-based pricing framework was developed to recommend premium adjustments.

Instead of charging nearly identical premiums to all policyholders, the proposed framework aligns premiums with expected insurance risk.

Current vs Recommended Premium

Risk Category	Current Premium (₹)	Recommended Premium (₹)
Very Low Risk	1,260	1,068
Low Risk	1,259	1,194
Medium Risk	1,259	1,256
High Risk	1,262	1,445
Very High Risk	1,217	1,633



(Figure: Current vs Recommended Premium Chart)

Pricing Insights

- Very Low Risk policyholders qualify for premium discounts.
- Medium Risk policyholders require minimal adjustment.
- High Risk and Very High Risk policyholders require higher premiums due to increased expected claim costs.
- The proposed framework improves pricing fairness by aligning premiums with historical claims experience.

Overall, the pricing model demonstrates the practical application of actuarial principles in developing data-driven insurance premium recommendations.

Interactive Power BI Dashboard

To present the analytical findings in an intuitive and business-friendly format, an interactive dashboard was developed using **Microsoft Power BI**.

The dashboard integrates claim frequency, claim severity, fraud analysis, risk segmentation, and pricing recommendations into a single reporting interface.

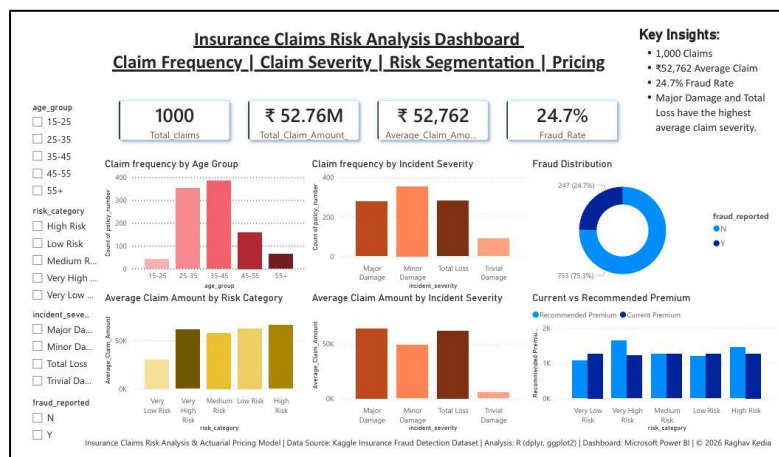


Figure: Insurance Claims Risk Analysis Dashboard

Dashboard Features

- Portfolio KPIs
 - Claim Frequency Analysis
 - Claim Severity Analysis
 - Fraud Analysis
 - Risk Category Distribution
 - Current vs Recommended Premium Comparison
 - Interactive Filters and Slicers
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Key Dashboard Insights

- Collision-related incidents account for the majority of insurance claims.
- Major Damage incidents generate the highest claim costs.
- Fraudulent claims exhibit higher average claim amounts.
- Very Low Risk policyholders experience substantially lower claim costs.
- Premium recommendations increase progressively with policyholder risk.

The dashboard provides insurers with an effective decision-support tool for monitoring portfolio performance and evaluating pricing strategies.

Conclusion

This project demonstrates how actuarial principles and modern data analytics techniques can be combined to evaluate insurance claims and support data-driven pricing decisions.

Using **1,000 insurance claim records** and **39 variables**, the project explored claim behaviour through exploratory data analysis, claim frequency analysis, claim severity analysis, fraud assessment, and risk segmentation. A five-level rule-based risk segmentation framework was developed to classify policyholders according to their expected insurance risk.

Building on this segmentation, a risk-based pricing framework was proposed to recommend differentiated premiums based on historical claims experience. The results illustrate how insurers can improve pricing fairness and underwriting decisions by aligning premiums with expected claim costs.

An interactive Power BI dashboard further enhanced the project by presenting key findings through dynamic visualizations and business intelligence reporting. Overall, the project highlights the value of integrating **R, Power BI, Excel, and actuarial thinking** to transform historical insurance data into actionable business insights.